

Lemniscate acceleration: a self-contained convergence proof via performance estimation

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July 2026

1 Introduction

The *lemniscate accelerated gradient method* (LemniAcc) is a fixed-step first-order method for minimizing an L -smooth convex function $f: \mathbb{R}^d \rightarrow \mathbb{R}$. After N gradient evaluations it returns a point x_N with

$$\|\nabla f(x_N)\|^2 \leq \frac{L^2 \|x_0 - x_\star\|^2}{\Omega_N^2} \leq \frac{\varpi^4 L^2 \|x_0 - x_\star\|^2}{(N+1)^4},$$

where $x_\star$ is a minimizer of f , the constant Ω_N is defined by a scalar recurrence, and $\varpi = 2 \int_0^1 dx / \sqrt{1-x^4}$ is the lemniscate constant. This document is a self-contained supplement to the companion paper [1], where the method is introduced: nothing here depends on the companion paper, and the companion paper also contains a Lyapunov-function proof of the same theorem, which is not reproduced here.

The proof strategy of this document is the performance estimation (PEP) methodology [3, 11]. We formulate the worst-case performance of the method as an optimization problem, relax it to a convex semidefinite program (SDP), pass to the dual SDP, and then exhibit an explicit analytic feasible point of that dual whose objective value is $1/\Omega_N^2$. Weak duality converts this feasible point into the convergence guarantee. Only elementary linear algebra and one-dimensional calculus are used; in particular, strong duality is never invoked.

The document is organized as follows. Section 2 fixes notation. Section 3 states the algorithm, its parameter recurrence, and the convergence theorem. Section 4 proves that the parameters exist, are unique, are computable by bisection, and satisfy the bounds used in the theorem. Section 5 develops the PEP formulation and its dual. Section 6 states the analytic certificate, and Section 7 verifies its feasibility, completing the proof of the convergence theorem for the h -matrix form of the method. Section 8 proves that the h -matrix form coincides with LemniAcc. Section 9 records how the certificate was found.

2 Notation

Write $\mathbb{R}^d$ for the underlying Euclidean space and write $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ for the standard inner product and norm on $\mathbb{R}^d$. Write $\mathbb{R}^{m \times n}$ for the set of $m \times n$ matrices, $\mathbb{S}^n$ for the set of $n \times n$ symmetric matrices, and $\mathbb{S}_+^n$ for the set of $n \times n$ positive-semidefinite matrices. We use the standard notation e_i for the unit vector having a single 1 as its i -th component. Write $(\cdot \odot \cdot) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{S}^n$ to denote the symmetric outer product, that is, for any $x, y \in \mathbb{R}^n$, $x \odot y = (xy^\top + yx^\top)/2$, and $\delta_{i,j}$ denotes the Kronecker delta, equal to 1 if $i = j$ and 0 otherwise. We write $\mathbf{tr}$ for the trace of a square matrix.

For $L > 0$, we write $\mathcal{F}_{0,L}$ for the class of L -smooth convex functions $f: \mathbb{R}^d \rightarrow \mathbb{R}$, that is, differentiable convex functions for which ∇f is L -Lipschitz. Throughout, x_* denotes a minimizer of the function f under consideration and $f_* \triangleq f(x_*)$. For a method that runs for N iterations and queries the points $x_0, x_1, \dots, x_N$, we collect the iteration indices together with the optimal point in the index set $I_N^* \triangleq \{0, 1, \dots, N, *\}$. Finally, we use the single-gradient-step notation

$$z^+ \triangleq z - \frac{1}{L} \nabla f(z), \quad \forall z \in \mathbb{R}^d.$$

3 The algorithm and its guarantee

Consider the unconstrained minimization problem

$$\underset{x \in \mathbb{R}^d}{\text{minimize}} \quad f(x), \tag{1}$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is an L -smooth convex function. We seek an algorithm that uses N gradient evaluations to produce an output x_N with a small worst-case guarantee on $\|\nabla f(x_N)\|$.

Algorithm statement. The lemniscate accelerated gradient method is

$$\begin{aligned} z_{k+1} &= z_k - \Omega_N \left(\frac{1 + \rho_{k+1}^2}{2\rho_{k+1}} - \frac{1 + \rho_k^2}{2\rho_k} \right) \frac{1}{L} \nabla f(x_k) \\ x_{k+1} &= x_k^+ + \left(\frac{1 + \rho_{k+1}^2}{1 - \rho_{k+1}^2} - \frac{1 + \rho_{k+2}^2}{1 - \rho_{k+2}^2} \right) z_{k+1} \end{aligned} \tag{LemniAcc}$$

for $k = 0, 1, 2, \dots, N-1$, where N is a prespecified iteration count, $x_0 \in \mathbb{R}^d$ is the starting point, $z_0 = 0$, and $\rho_0, \rho_1, \dots, \rho_N, \rho_{N+1}$ and Ω_N are positive constants that we define in Lemma 1.

We can also express LemniAcc equivalently in the so-called momentum form

$$\begin{aligned} x_1 &= x_0^+ + \alpha^{(\text{init})}(x_0^+ - x_0), \\ x_{k+1} &= x_k^+ + \alpha_k^{(\text{mom})}(x_k^+ - x_{k-1}^+) + \alpha_k^{(\text{corr})}(x_k^+ - x_k), \quad k = 1, \dots, N-1, \end{aligned} \tag{2}$$

where

$$\begin{aligned} \alpha^{(\text{init})} &= \Omega_N \left(\frac{1 + \rho_1^2}{2\rho_1} - \frac{1 + \rho_0^2}{2\rho_0} \right) \left(\frac{1 + \rho_N^2}{2\rho_N} - \frac{1 + \rho_{N-1}^2}{2\rho_{N-1}} \right), \\ \alpha_k^{(\text{mom})} &= \frac{\frac{1 + \rho_{N-k}^2}{2\rho_{N-k}} - \frac{1 + \rho_{N-k-1}^2}{2\rho_{N-k-1}}}{\frac{1 + \rho_{N-k+1}^2}{2\rho_{N-k+1}} - \frac{1 + \rho_{N-k}^2}{2\rho_{N-k}}}, \\ \alpha_k^{(\text{corr})} &= \left(\frac{1 + \rho_{N-k}^2}{2\rho_{N-k}} - \frac{1 + \rho_{N-k-1}^2}{2\rho_{N-k-1}} \right) \left[\Omega_N \left(\frac{1 + \rho_{k+1}^2}{2\rho_{k+1}} - \frac{1 + \rho_k^2}{2\rho_k} \right) - \frac{1}{\frac{1 + \rho_{N-k+1}^2}{2\rho_{N-k+1}} - \frac{1 + \rho_{N-k}^2}{2\rho_{N-k}}} \right]. \end{aligned}$$

The two forms are equivalent in the sense that they produce the same iterates $x_0, \dots, x_N$, by Lemma 7 of Section 8.

N	Ω_N	ρ_1	ρ_2	ρ_3	ρ_4	ρ_5	ρ_6	ρ_7
1	2.4142	0.4142						
2	4.1628	0.6126	0.2402					
3	6.2479	0.7241	0.4142	0.1601				
4	8.6645	0.7931	0.5380	0.3004	0.1154			
5	11.4075	0.8388	0.6277	0.4142	0.2287	0.0877		
6	14.4730	0.8707	0.6942	0.5052	0.3287	0.1805	0.0691	
7	17.8578	0.8939	0.7446	0.5780	0.4142	0.2675	0.1464	0.0560

Table 1: Values of Ω_N and $\rho_1, \dots, \rho_N$ for $N = 1, \dots, 7$, computed by the bisection method.

Construction of the coefficients. To fully define LemniAcc, it remains to define the coefficients $\rho_0, \rho_1, \dots, \rho_N, \rho_{N+1}$ and Ω_N .

Lemma 1. *There exists a unique $\Omega_N > 0$ for which there is a strictly decreasing nonnegative sequence $1 = \rho_0 > \rho_1 > \dots > \rho_{N+1} = 0$ satisfying the recurrence equation*

$$\Omega_N (\rho_k - \rho_{k+1})^2 = \rho_k (1 - \rho_{k+1}^2), \quad k = 0, \dots, N. \quad (3)$$

Moreover, for this value of Ω_N , the sequence $\rho_0, \rho_1, \dots, \rho_{N+1}$ is uniquely determined.

Lemma 1 is proved in Section 4. We can calculate the numerical values of $\{\rho_k\}_{k=0}^{N+1}$ and Ω_N using bisection based on $1 \leq \Omega_N \leq (N+1)^2$: we start with $\Omega_l = 1$ and $\Omega_r = (N+1)^2$. For a candidate $\Omega_m = (\Omega_l + \Omega_r)/2$, we start from $\rho_0 = 1$ and use the recurrence (3) to calculate the subsequent values of ρ_n . If $\rho_n < 1/\Omega_m$ for some $n \leq N$, we increase the value of Ω (set $\Omega_l \leftarrow \Omega_m$). If $\rho_{N+1} > 0$, we decrease the value of Ω (set $\Omega_r \leftarrow \Omega_m$). Continuing this method, $\Omega_r - \Omega_l$ halves at each step, and $\Omega_m \rightarrow \Omega_N$, so we can calculate Ω_N to arbitrary precision. Corollary 1 proves that both update rules are valid and that the initial bracket always contains Ω_N , so the bisection indeed converges to Ω_N . Table 1 reports the values of Ω_N and of the nontrivial entries $\rho_1, \dots, \rho_N$ for $N = 1, \dots, 7$.

We note that the sequence $\{\rho_k\}_{k=0}^{N+1}$ has a symmetry: the map $T(\rho) \triangleq (1 - \rho)/(1 + \rho)$, which satisfies $T \circ T = \text{id}$, relates mirrored entries through $\rho_{N+1-k} = T(\rho_k)$ for $0 \leq k \leq N+1$. This follows from the uniqueness in Lemma 1 together with the fact that $(\rho_k, \rho_{k+1}) = (a, b)$ satisfies the recurrence (3) if and only if $(T(b), T(a))$ does. In particular, for odd N , the middle index $k = (N+1)/2$ is its own mirror image, so $\rho_{(N+1)/2}$ must be the point such that $T(\rho) = \rho$, namely $\rho_{(N+1)/2} = \sqrt{2} - 1 \approx 0.4142$.

Convergence theorem. The main result of this document is the following guarantee.

Theorem 1. *Let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be an L -smooth convex function with a minimizer x_* . Then, the final iterate x_N of (LemniAcc) satisfies*

$$\|\nabla f(x_N)\|^2 \leq \frac{L^2 \|x_0 - x_*\|^2}{\Omega_N^2} \leq \frac{\varpi^4 L^2 \|x_0 - x_*\|^2}{(N+1)^4}.$$

Here, $\varpi = 2 \int_0^1 1/\sqrt{1-x^4} dx$ is the lemniscate constant, and the latter relaxation comes from $(N+1)^2/\varpi^2 < \Omega_N$ (Corollary 1). The proof of the first inequality occupies Sections 5–8.

LemniAcc is not the first method to attain an $O(L^2 \|x_0 - x_*\|^2 / N^4)$ rate on the squared gradient norm $\|\nabla f(x_N)\|^2$ of L -smooth convex functions. Nesterov et al. [10, Remark 2.1] observed that

running Nesterov's accelerated gradient method (AGM) [9] (or Kim–Fessler's optimized gradient method (OGM) [5]) for the first half of the iterations and OGM-G [6] for the second half achieves this rate, which matches the $\Omega(L^2 \|x_0 - x_\star\|^2 / N^4)$ lower bound of [7, 8] and is therefore optimal up to a constant factor.

Lemniscate acceleration improves upon this two-phase scheme in two ways. First, it attains the optimal rate as a single method, with one coefficient sequence and no switch between two algorithms. Second, it improves the constant of the guarantee. For even N , chaining the standard guarantee of OGM with that of OGM-G yields

$$\|\nabla f(x_N)\|^2 \leq \frac{64 L^2 \|x_0 - x_\star\|^2}{(N+2)^4}.$$

In comparison, Theorem 1 attains the constant $\varpi^4 \approx 47.27$, improving the constant by 1.35.

4 Construction of the parameters

In this section we prove Lemma 1, establish the bounds on Ω_N used in Theorem 1, and prove that the bisection of Section 3 is correct.

The one-step map. For $\Omega > 0$ and $\rho \in (0, 1]$, a pair of consecutive terms of (3) solves

$$\Omega(\rho - t)^2 = \rho(1 - t^2) \tag{4}$$

in the unknown t , which rearranges to the quadratic equation

$$(\Omega + \rho)t^2 - 2\Omega\rho t + \Omega\rho^2 - \rho = 0.$$

We work on the admissible set

$$\mathcal{A} \triangleq \{(\Omega, \rho) : \Omega > 0, 0 < \rho \leq 1, \Omega\rho \geq 1\}.$$

Lemma 2 (One-step map). *For $(\Omega, \rho) \in \mathcal{A}$, equation (4) has exactly one solution t with $0 \leq t < \rho$, namely*

$$F_\Omega(\rho) \triangleq \frac{\Omega\rho - \sqrt{\rho^2 + \Omega\rho(1 - \rho^2)}}{\Omega + \rho}, \tag{5}$$

and the other solution is greater than or equal to ρ , with equality only when $\rho = 1$. Moreover:

1. $F_\Omega(\rho) = 0$ if and only if $\Omega\rho = 1$;
2. F is continuous on $\mathcal{A}$ and strictly increasing in each of Ω and ρ on the fibers of $\mathcal{A}$ (the other variable being fixed).

Proof. The quarter-discriminant of the quadratic equals $\Omega^2\rho^2 - (\Omega + \rho)(\Omega\rho^2 - \rho) = \rho^2 + \Omega\rho(1 - \rho^2) > 0$, so (4) has the two real solutions

$$t_\pm = \frac{\Omega\rho \pm \sqrt{\rho^2 + \Omega\rho(1 - \rho^2)}}{\Omega + \rho}.$$

Since $\Omega\rho(1 - \rho^2) \geq 0$, the square root is at least ρ , hence

$$t_+ \geq \frac{\Omega\rho + \rho}{\Omega + \rho} \geq \frac{\Omega\rho + \rho^2}{\Omega + \rho} = \rho,$$

and both inequalities hold with equality only when $\rho = 1$. On the other hand, $t_- < \Omega\rho/(\Omega + \rho) < \rho$. The product of the two solutions is $t_+t_- = \rho(\Omega\rho - 1)/(\Omega + \rho) \geq 0$ on $\mathcal{A}$; since $t_+ \geq \rho > 0$, this gives $t_- \geq 0$, with $t_- = 0$ exactly when $\Omega\rho = 1$. Therefore $F_\Omega(\rho) = t_-$ is the unique solution in $[0, \rho)$, which proves the root dichotomy and claim 1.

For claim 2, continuity follows from (5) because the denominator is positive. For the strict monotonicity, define

$$\psi(\Omega, \rho, t) \triangleq \Omega(\rho - t)^2 - \rho(1 - t^2),$$

and fix a point in the interior of $\mathcal{A}$ with $t = F_\Omega(\rho) \in [0, \rho)$. Substituting the root identity $1 - t^2 = \Omega(\rho - t)^2/\rho$ into the partial derivatives of ψ and using $0 \leq t < \rho \leq 1$, tedious but straightforward algebra gives $\psi_\Omega > 0$, $\psi_\rho > 0$, and $\psi_t < 0$ at this point. By the implicit function theorem, F is continuously differentiable in the interior of $\mathcal{A}$ with $\partial F/\partial \Omega = -\psi_\Omega/\psi_t > 0$ and $\partial F/\partial \rho = -\psi_\rho/\psi_t > 0$. Consequently, for fixed $\rho \in (0, 1)$ the map $\Omega \mapsto F_\Omega(\rho)$ is continuous on $[1/\rho, \infty)$ and strictly increasing on its interior, hence strictly increasing on $[1/\rho, \infty)$; the same argument applies to $\rho \mapsto F_\Omega(\rho)$ on $[1/\Omega, 1]$ for fixed $\Omega > 1$; and on the remaining boundary segment $\rho = 1$ the explicit formula $F_\Omega(1) = (\Omega - 1)/(\Omega + 1)$ is strictly increasing in Ω . $\square$

Shooting orbits. For $\Omega \geq 1$ define the shooting iterates

$$\Phi_0(\Omega) \triangleq 1, \quad \Phi_{i+1}(\Omega) \triangleq F_\Omega(\Phi_i(\Omega)),$$

defined for as long as $(\Omega, \Phi_i(\Omega)) \in \mathcal{A}$. Taking the positive square root in (4) gives the step identity

$$\Phi_i(\Omega) - \Phi_{i+1}(\Omega) = \sqrt{\frac{\Phi_i(\Omega)(1 - \Phi_{i+1}(\Omega)^2)}{\Omega}} \leq \frac{1}{\sqrt{\Omega}} \quad (6)$$

whenever the left-hand side is defined; the final bound holds because the iterates satisfy $0 \leq \Phi_{i+1}(\Omega) < \Phi_i(\Omega) \leq \Phi_0(\Omega) = 1$ (Lemma 2), so $\Phi_i(\Omega)(1 - \Phi_{i+1}(\Omega)^2) \leq 1$.

Proof of Lemma 1. Call a pair $(\Omega, \{\rho_i\}_{i=0}^{N+1})$ *admissible for horizon N* if it satisfies all requirements of the lemma: $1 = \rho_0 > \rho_1 > \dots > \rho_{N+1} = 0$ and (3) for $0 \leq i \leq N$.

Step 0: every admissible pair is a shooting orbit. Let $(\Omega, \{\rho_i\})$ be admissible for horizon N . Strict decrease gives $\rho_i > \rho_{N+1} = 0$ for $i \leq N$. The recurrence at $i = N$ reads $\Omega\rho_N^2 = \rho_N$, so $\rho_N = 1/\Omega$, and the decrease of the sequence gives $\Omega\rho_i \geq \Omega\rho_N = 1$ for $i \leq N$; hence $(\Omega, \rho_i) \in \mathcal{A}$ for $0 \leq i \leq N$. In particular $1/\Omega = \rho_N \leq \rho_0 = 1$ forces $\Omega \geq 1$, and for $N \geq 1$ the strict inequality $\rho_N < \rho_0$ forces $\Omega > 1$. Finally, for $0 \leq i \leq N$ the term ρ_{i+1} solves (4) with $0 \leq \rho_{i+1} < \rho_i$, so Lemma 2 gives $\rho_{i+1} = F_\Omega(\rho_i)$. Therefore $\rho_i = \Phi_i(\Omega)$ for all $0 \leq i \leq N+1$: the sequence is uniquely determined by Ω , which is the final claim of the lemma.

Step 1: existence, by induction on the horizon. We construct numbers $1 = \Omega_0 < \Omega_1 < \Omega_2 < \dots$ such that $(\Omega_N, \{\Phi_i(\Omega_N)\}_{i=0}^{N+1})$ is admissible for horizon N . For $N = 0$ the requirements reduce to $\Omega(\rho_0 - \rho_1)^2 = \rho_0(1 - \rho_1^2)$ with $\rho_0 = 1$ and $\rho_1 = 0$, that is, $\Omega_0 = 1$; and indeed $\Phi_1(1) = F_1(1) = 0$.

For the induction step, assume $(\Omega_N, \{\Phi_i(\Omega_N)\})$ is admissible for horizon N . We first record, by induction on $i \in \{1, \dots, N+1\}$, that the maps $\Omega \mapsto \Phi_i(\Omega)$ are well defined, continuous, and strictly increasing on the ray $[\Omega_N, \infty)$. For $i = 1$, $\Phi_1(\Omega) = F_\Omega(1) = (\Omega - 1)/(\Omega + 1)$ has all three properties. For the step from i to $i+1$ (with $i \leq N$), the monotonicity of F and admissibility at horizon N give the domain bound

$$\Omega \Phi_i(\Omega) \geq \Omega_N \Phi_i(\Omega_N) \geq \Omega_N \Phi_N(\Omega_N) = \Omega_N \cdot \frac{1}{\Omega_N} = 1,$$

so $(\Omega, \Phi_i(\Omega)) \in \mathcal{A}$ and $\Phi_{i+1}(\Omega) = F_\Omega(\Phi_i(\Omega))$ is well defined; it is continuous by Lemma 2 and the induction hypothesis, and strictly increasing because, for $\Omega_N \leq \Omega < \Omega'$,

$$\Phi_{i+1}(\Omega) = F_\Omega(\Phi_i(\Omega)) < F_{\Omega'}(\Phi_i(\Omega)) < F_{\Omega'}(\Phi_i(\Omega')) = \Phi_{i+1}(\Omega'),$$

the middle application being legitimate since $\Omega' \Phi_i(\Omega) > \Omega \Phi_i(\Omega) \geq 1$, and the second inequality being strict by the induction hypothesis $\Phi_i(\Omega) < \Phi_i(\Omega')$.

Now consider $h(\Omega) \triangleq \Omega \Phi_{N+1}(\Omega)$ on $[\Omega_N, \infty)$. It is continuous, and it is strictly increasing because Φ_{N+1} is nonnegative and strictly increasing there: for $\Omega_N \leq \Omega < \Omega'$, $\Omega \Phi_{N+1}(\Omega) \leq \Omega \Phi_{N+1}(\Omega') < \Omega' \Phi_{N+1}(\Omega')$. Moreover $h(\Omega_N) = \Omega_N \cdot \Phi_{N+1}(\Omega_N) = 0$, and telescoping (6) over $0 \leq i \leq N$, all iterates being defined on this ray, gives $\Phi_{N+1}(\Omega) \geq 1 - (N+1)/\sqrt{\Omega}$, so $h(\Omega) \geq \Omega - (N+1)\sqrt{\Omega} \rightarrow \infty$ as $\Omega \rightarrow \infty$. By the intermediate value theorem there is a unique $\Omega_{N+1} \in (\Omega_N, \infty)$ with $h(\Omega_{N+1}) = 1$, that is,

$$\Phi_{N+1}(\Omega_{N+1}) = \frac{1}{\Omega_{N+1}}.$$

The sequence $\rho_i \triangleq \Phi_i(\Omega_{N+1})$ for $0 \leq i \leq N+1$, extended by $\rho_{N+2} \triangleq 0$, is then admissible for horizon $N+1$: the recurrence holds for $0 \leq i \leq N$ by construction, and at $i = N+1$ it reads $\Omega_{N+1}(\rho_{N+1} - 0)^2 = 1/\Omega_{N+1} = \rho_{N+1}(1 - 0^2)$; the sequence is strictly decreasing because $F_\Omega(\rho) < \rho$ at every step and $\rho_{N+1} = 1/\Omega_{N+1} > 0 = \rho_{N+2}$; and it is nonnegative because $\rho_i \geq \rho_{N+1} > 0$ for $i \leq N+1$. This completes the induction, so an admissible Ω exists for every horizon.

Step 2: uniqueness of Ω . Suppose $(\Omega, \{\rho_i\})$ and $(\Omega', \{\rho'_i\})$ are both admissible for horizon N with $\Omega < \Omega'$. By Step 0, both sequences are shooting orbits, every pair (Ω, ρ_i) and (Ω', ρ'_i) with $i \leq N$ lies in $\mathcal{A}$, and $\Omega, \Omega' \geq 1$. For $N = 0$, Step 0 gives $\rho_1 = (\Omega - 1)/(\Omega + 1) = 0$ and $\rho'_1 = (\Omega' - 1)/(\Omega' + 1) = 0$, forcing $\Omega = \Omega' = 1$ and contradicting $\Omega < \Omega'$; so assume $N \geq 1$. We prove $\rho_i < \rho'_i$ for $1 \leq i \leq N+1$ by induction. For $i = 1$,

$$\rho_1 = \frac{\Omega - 1}{\Omega + 1} < \frac{\Omega' - 1}{\Omega' + 1} = \rho'_1.$$

For the induction step, the strict monotonicity of F in both arguments (Lemma 2) gives

$$\rho_{i+1} = F_\Omega(\rho_i) < F_{\Omega'}(\rho_i) < F_{\Omega'}(\rho'_i) = \rho'_{i+1},$$

where the middle steps are valid because $\Omega' \rho_i > \Omega \rho_i \geq 1$, and the second inequality is strict by the induction hypothesis $\rho_i < \rho'_i$. Iterating the induction through the final step $i = N$ yields $\rho_{N+1} < \rho'_{N+1}$, contradicting $\rho_{N+1} = \rho'_{N+1} = 0$. Hence the admissible Ω is unique, and Step 0 shows the sequence is uniquely determined by it. $\square$

The same machinery yields the quantitative bounds on Ω_N quoted in Theorem 1 and the correctness of the bisection described in Section 3.

Corollary 1 (Bounds on Ω_N and bisection correctness). *Let $N \geq 1$, and let Ω_N and $\{\rho_i\}_{i=0}^{N+1}$ be as in Lemma 1. Then:*

1. $\frac{(N+1)^2}{\varpi^2} < \Omega_N < (N+1)^2$, where ϖ is the lemniscate constant defined in Theorem 1; moreover $1 = \Omega_0 < \Omega_1 < \Omega_2 < \dots$.
2. Fix $\Omega \geq 1$ and generate the shooting iterates $\Phi_0(\Omega), \Phi_1(\Omega), \dots$, computing $\Phi_{i+1}(\Omega) = F_\Omega(\Phi_i(\Omega))$ whenever $\Phi_i(\Omega) \geq 1/\Omega$. If $\Omega > \Omega_N$, then $\Phi_i(\Omega) > 1/\Omega$ for all $0 \leq i \leq N$ and $\Phi_{N+1}(\Omega) > 0$. If $\Omega < \Omega_N$, then there is an index $n \leq N$ with $\Phi_n(\Omega) < 1/\Omega$. If $\Omega = \Omega_N$, the iterates realize the sequence of Lemma 1. Consequently both update rules of the bisection in Section 3 are correct, the initial bracket $[\Omega_l, \Omega_r] = [1, (N+1)^2]$ contains Ω_N , the invariant $\Omega_l < \Omega_N \leq \Omega_r$ is maintained, and the midpoints converge to Ω_N .

Proof. Part 1. At $\Omega = \Omega_N$ the step identity (6) holds for $0 \leq i \leq N$, and telescoping gives

$$1 = \rho_0 - \rho_{N+1} = \sum_{i=0}^N \sqrt{\frac{\rho_i (1 - \rho_{i+1}^2)}{\Omega_N}} \leq \frac{N+1}{\sqrt{\Omega_N}},$$

and the term $i = 1$, present since $N \geq 1$, is strictly smaller than $1/\sqrt{\Omega_N}$ because $\rho_1 < 1$; hence $\sqrt{\Omega_N} < N+1$. For the lower bound, dividing the step identity by $\sqrt{\rho_i(1 - \rho_{i+1}^2)}$ and summing gives

$$\frac{N+1}{\sqrt{\Omega_N}} = \sum_{i=0}^N \frac{\rho_i - \rho_{i+1}}{\sqrt{\rho_i (1 - \rho_{i+1}^2)}} = \sum_{i=0}^N \int_{\rho_{i+1}}^{\rho_i} \frac{dt}{\sqrt{\rho_i (1 - t^2)}} < \sum_{i=0}^N \int_{\rho_{i+1}}^{\rho_i} \frac{dt}{\sqrt{t(1 - t^2)}} = \int_0^1 \frac{dt}{\sqrt{t(1 - t^2)}} = \varpi,$$

where the strict inequality holds because on the interior of each interval the integrand on the right strictly dominates the constant integrand on the left, and the substitution $t = s^2$ identifies the comparison integral with the form $\varpi = 2 \int_0^1 ds / \sqrt{1 - s^4}$ stated after Theorem 1. Hence $\Omega_N > (N+1)^2/\varpi^2$. Strict monotonicity of $N \mapsto \Omega_N$ was established in Step 1 of the proof of Lemma 1 and is inherited by the unique admissible values.

Part 2. If $\Omega > \Omega_N$, the sub-induction in Step 1 (anchored at Ω_N) shows that $\Phi_i(\Omega)$ is defined and strictly increasing in Ω for $1 \leq i \leq N+1$; therefore, for $0 \leq i \leq N$, using that the iterates decrease in the index ($F_\Omega(\rho) < \rho$),

$$\Phi_i(\Omega) \geq \Phi_N(\Omega) > \Phi_N(\Omega_N) = \frac{1}{\Omega_N} > \frac{1}{\Omega},$$

and $\Phi_{N+1}(\Omega) > \Phi_{N+1}(\Omega_N) = 0$. If $1 \leq \Omega < \Omega_N$, suppose toward a contradiction that $\Phi_i(\Omega) \geq 1/\Omega$ for all $0 \leq i \leq N$; then all iterates up to $\Phi_N(\Omega)$ are defined. The comparison induction of Step 2 does not use admissibility of the lower orbit at the terminal index: it only needs both shooting orbits to be defined up to the index being compared, the domain of each step being guaranteed here by the contradiction hypothesis and by $\Omega_N \Phi_i(\Omega) > \Omega \Phi_i(\Omega) \geq 1$. The same induction therefore gives $\Phi_N(\Omega) < \Phi_N(\Omega_N) = 1/\Omega_N < 1/\Omega$, a contradiction. The case $\Omega = \Omega_N$ is Lemma 1 itself. For the bisection, Part 1 gives $\Omega_l = 1 < \Omega_N < (N+1)^2 = \Omega_r$ at initialization, and by the sign characterization just proved each update rule fires exactly on its side of Ω_N (neither fires when $\Omega_m = \Omega_N$). Hence each update preserves $\Omega_l < \Omega_N \leq \Omega_r$ while halving the bracket, and the midpoints converge to Ω_N . $\square$

The comparison with the integral in Part 1 in fact identifies the limit of $\Omega_N/(N+1)^2$: the lower bound of Corollary 1 is asymptotically tight.

Lemma 3. *Let Ω_N be as in Lemma 1. Then*

$$\lim_{N \rightarrow \infty} \frac{\Omega_N}{(N+1)^2} = \frac{1}{\varpi^2}.$$

Proof. Write $\{\rho_i\}_{i=0}^{N+1}$ for the sequence of Lemma 1 at Ω_N , put $\Delta_i \triangleq \rho_i - \rho_{i+1} > 0$, and set

$$g(x) \triangleq \frac{1}{\sqrt{x(1-x^2)}}, \quad 0 < x < 1,$$

so that $\int_0^1 g(x) dx = \varpi$, as recorded in the proof of Corollary 1. As in Part 1 of that proof, dividing the step identity (6) at $\Omega = \Omega_N$ by $\sqrt{\rho_i(1 - \rho_{i+1}^2)}$ and summing over $0 \leq i \leq N$ gives

$$\frac{N+1}{\sqrt{\Omega_N}} = \sum_{i=0}^N \frac{\Delta_i}{\sqrt{\rho_i(1 - \rho_{i+1}^2)}}. \quad (7)$$

We claim that the right-hand side converges to ϖ ; the lemma follows, since then $\sqrt{\Omega_N}/(N+1) \rightarrow 1/\varpi$.

The intervals $[\rho_{i+1}, \rho_i]$, $0 \leq i \leq N$, partition $[0, 1]$, and by the step identity and the lower bound of Corollary 1 their lengths satisfy

$$\delta_N \triangleq \max_{0 \leq i \leq N} \Delta_i \leq \frac{1}{\sqrt{\Omega_N}} < \frac{\varpi}{N+1}, \quad (8)$$

so the mesh δ_N tends to zero. For $x \in [\rho_{i+1}, \rho_i]$ we have $x \leq \rho_i$ and $1 - x^2 \leq 1 - \rho_{i+1}^2$, hence $g(x) \geq 1/\sqrt{\rho_i(1 - \rho_{i+1}^2)}$, so the right-hand side of (7) is a lower Riemann-type sum for $\int_0^1 g(x) dx = \varpi$. Because g is integrable on $(0, 1)$ and the two-variable comparison kernel $(x, y) \mapsto 1/\sqrt{x(1 - y^2)}$ is uniformly continuous on compact squares inside $(0, 1)^2$, a routine argument (splitting off small neighborhoods of the endpoints and comparing termwise on the interior, tedious but straightforward) shows that the sum converges to ϖ as the mesh tends to zero. Hence $(N+1)/\sqrt{\Omega_N} \rightarrow \varpi$. $\square$

5 PEP formulation of the worst case

We now formulate the worst-case performance of a fixed-step first-order method as an optimization problem and reduce it to a convex SDP whose dual carries convergence certificates. This is the performance estimation methodology of Drori and Teboulle [3] in the exact form given by Taylor, Hendrickx, and Glineur [11].

A *fixed-step first-order method* (FSFOM) with N steps produces its iterates from a starting point $x_0 \in \mathbb{R}^d$ with

$$x_i = x_{i-1} - \frac{1}{L} \sum_{j=0}^{i-1} h_{i,j} \nabla f(x_j)$$

for $1 \leq i \leq N$, where the stepsizes $\{h_{i,j}\}_{0 \leq j < i \leq N}$ are fixed in advance, normalized here by L . We can equivalently express the FSFOM as

$$x_i = x_0 - \frac{1}{L} \sum_{j=0}^{i-1} \alpha_{i,j} \nabla f(x_j),$$

where $\alpha_{i,j} \triangleq \sum_{k=j+1}^i h_{k,j}$ for $0 \leq j < i \leq N$; this triangular relationship between h and α is invertible, so we can compute h given α and vice versa. The stepsizes h or α may depend on $\mathcal{F}_{0,L}$ and the value of N , but are otherwise predetermined.

For our setup, the function class is $\mathcal{F}_{0,L}$, the performance measure is $\|\nabla f(x_N)\|^2$, and the initial condition is $\|x_0 - x_\star\|^2 \leq R^2$ for a given $R > 0$. The *worst-case performance* $\mathcal{R}(M)$ of an FSFOM

M is then the optimal value of

$$\mathcal{R}(M) = \left(\begin{array}{l} \text{maximize} \quad \|\nabla f(x_N)\|^2 \\ \text{subject to} \\ f \in \mathcal{F}_{0,L}, \\ x_\star \text{ is an optimal solution to (1) satisfying } \nabla f(x_\star) = 0, \\ \{x_i\}_{1 \leq i \leq N} \text{ is generated by } M \text{ with initial point } x_0, \\ \|x_0 - x_\star\|^2 \leq R^2, \\ x_\star = 0, f(x_\star) = 0 \end{array} \right), \quad (\mathcal{O}^{\text{inner}})$$

where $f, x_0, \dots, x_N$, and $x_\star$ are the decision variables. We set $x_\star = 0$ and $f(x_\star) = 0$ without any loss of generality because the function class $\mathcal{F}_{0,L}$ is closed and invariant under shifting variables and function values. Any upper bound on $\mathcal{R}(M)$ is a convergence guarantee for M ; producing such a bound for the stepsizes of LemniAcc is exactly what the rest of this document does.

We reduce $(\mathcal{O}^{\text{inner}})$ to a finite-dimensional convex SDP in two steps: we first discretize the function variable through an interpolation theorem, and we then lift the resulting finite-dimensional problem to a Gram-matrix formulation.

As the first step, we use a discretization result [11, Corollary 1], which states that in the inner problem we can discretize f at the points seen by the method without any loss of generality.

Theorem 2 ($\mathcal{F}_{0,L}$ -interpolation [11, Corollary 1]). *Let I be an index set, and let $\{(x_i, g_i, f_i)\}_{i \in I} \subseteq \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}$. Let $L > 0$. There exists $f \in \mathcal{F}_{0,L}$ satisfying $f(x_i) = f_i$ and $g_i \in \partial f(x_i)$ for all $i \in I$ if and only if*

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2$$

for all $i, j \in I$.

In other words, we can discretize the cocoercivity at the points $x_0, x_1, \dots, x_N, x_\star$ by using Theorem 2 and by introducing the notation $g_i \triangleq \nabla f(x_i)$ and $f_i \triangleq f(x_i)$. Thus, we cast $(\mathcal{O}^{\text{inner}})$ as the following finite-dimensional but nonconvex problem:

$$\mathcal{R}(M) = \left(\begin{array}{l} \text{maximize} \quad \|g_N\|^2 \\ \text{subject to} \quad f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2, \quad i, j \in I_N^\star, \\ g_\star = 0, \\ x_i = x_0 - \frac{1}{L} \sum_{j=0}^{i-1} \alpha_{i,j} g_j, \quad 1 \leq i \leq N, \\ \|x_0 - x_\star\|^2 \leq R^2, \\ x_\star = 0, f_\star = 0 \end{array} \right), \quad (9)$$

where the decision variables are $\{x_i, g_i, f_i\}_{i \in I_N^\star} \subseteq \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}$.

Next, we formulate (9) as a convex SDP. Let $P \triangleq [x_0 \mid g_0 \mid g_1 \mid \dots \mid g_N] \in \mathbb{R}^{d \times (N+2)}$, $G \triangleq P^\top P \in \mathbb{S}_+^{N+2}$, and $F \triangleq [f_0 \mid f_1 \mid \dots \mid f_N] \in \mathbb{R}^{1 \times (N+1)}$. Note that $\text{rank } G \leq d$. Define the following notation for selecting columns and elements of P and F :

$$\begin{aligned} \mathbf{g}_\star &= 0 \in \mathbb{R}^{N+2}, \quad \mathbf{g}_i = e_{i+2} \in \mathbb{R}^{N+2}, \quad 0 \leq i \leq N, \\ \mathbf{x}_0 &= e_1 \in \mathbb{R}^{N+2}, \quad \mathbf{x}_\star = 0 \in \mathbb{R}^{N+2}, \quad \mathbf{x}_i = \mathbf{x}_0 - \frac{1}{L} \sum_{j=0}^{i-1} \alpha_{i,j} \mathbf{g}_j \in \mathbb{R}^{N+2}, \quad 1 \leq i \leq N, \\ \mathbf{f}_\star &= 0 \in \mathbb{R}^{N+1}, \quad \mathbf{f}_i = e_{i+1} \in \mathbb{R}^{N+1}, \quad 0 \leq i \leq N. \end{aligned}$$

This notation is defined so that $x_i = P\mathbf{x}_i$, $g_i = P\mathbf{g}_i$, $f_i = F\mathbf{f}_i$ for $i \in I_N^*$. Note that $\mathbf{x}_i$ depends on $\{\alpha_{i,j}\}_{0 \leq j \leq i-1}$ linearly for $1 \leq i \leq N$. For $i, j \in I_N^*$, now define

$$\begin{aligned} A_{i,j}(\alpha) &= \mathbf{g}_j \odot (\mathbf{x}_i - \mathbf{x}_j) \in \mathbb{S}^{N+2}, \\ B_{i,j}(\alpha) &= (\mathbf{x}_i - \mathbf{x}_j) \odot (\mathbf{x}_i - \mathbf{x}_j) \in \mathbb{S}_+^{N+2}, \\ C_{i,j} &= (\mathbf{g}_i - \mathbf{g}_j) \odot (\mathbf{g}_i - \mathbf{g}_j) \in \mathbb{S}_+^{N+2}. \end{aligned} \quad (10)$$

Note that $A_{i,j}(\alpha)$ is affine and $B_{i,j}(\alpha)$ is quadratic as functions of $\{\alpha_{i,j}\}_{0 \leq j < i \leq N}$. Under this notation,

$$\begin{aligned} \langle g_j, x_i - x_j \rangle &= \mathbf{tr}GA_{i,j}(\alpha), \\ \|x_0 - x_\star\|^2 &= \mathbf{tr}GB_{0,\star}, \\ \|g_i - g_j\|^2 &= \mathbf{tr}GC_{i,j}, \end{aligned}$$

for $i, j \in I_N^*$. In these terms, (9) becomes a maximization over F and $G = P^\top P$, at the price of the rank constraint $\mathbf{rank} G \leq d$: the equivalence relies on the fact that any $G \in \mathbb{S}_+^{N+2}$ with $\mathbf{rank} G \leq d$ factors as $G = P^\top P$ with $P \in \mathbb{R}^{d \times (N+2)}$; see [11, §3.2]. We now drop the rank constraint. Dropping it can only enlarge the feasible set, so the optimal value of the resulting problem upper-bounds $\mathcal{R}(M)$ in every dimension d ; when $d \geq N + 2$ the rank constraint is vacuous and the relaxation is exact [11, §3.2]. Only the upper-bound direction is used in this document, so the guarantee we prove is dimension-free. The relaxed problem is the convex SDP

$$\mathcal{R}(M) \leq \left(\begin{array}{l} \text{maximize } \mathbf{tr}G(C_{N,\star}) \\ \text{subject to} \\ F(\mathbf{f}_j - \mathbf{f}_i) + \mathbf{tr}G[A_{i,j}(\alpha) + \frac{1}{2L}C_{i,j}] \leq 0, \quad i, j \in I_N^* : i \neq j, \quad \triangleright \text{dual var. } \lambda_{i,j} \geq 0 \\ -G \preceq 0, \quad \triangleright \text{dual var. } Z \succeq 0 \\ \mathbf{tr}GB_{0,\star} \leq R^2, \quad \triangleright \text{dual var. } \nu \geq 0 \end{array} \right), \quad (11)$$

where $F \in \mathbb{R}^{1 \times (N+1)}$ and $G \in \mathbb{S}^{N+2}$ are the decision variables. We denote the corresponding dual variables on the right hand side of the constraints for later use. Note that the right-hand side of (11) is free from the problem dimension d .

Next we use convex duality to formulate the SDP on the right-hand side of (11), originally a maximization problem, as a minimization problem. Its Lagrangian dual is

$$\overline{\mathcal{R}}(M) = \left(\begin{array}{l} \text{minimize } \nu R^2 \\ \text{subject to} \\ \sum_{i,j \in I_N^* : i \neq j} \lambda_{i,j} (\mathbf{f}_j - \mathbf{f}_i) = 0, \\ \nu B_{0,\star} - C_{N,\star} + \sum_{i,j \in I_N^* : i \neq j} \lambda_{i,j} [A_{i,j}(\alpha) + \frac{1}{2L}C_{i,j}] = Z, \\ Z \succeq 0, \\ \nu \geq 0, \lambda_{i,j} \geq 0, \quad i, j \in I_N^* : i \neq j \end{array} \right), \quad (12)$$

where $\nu \in \mathbb{R}$, $\lambda = \{\lambda_{i,j}\}_{i,j \in I_N^* : i \neq j}$, and $Z \in \mathbb{S}_+^{N+2}$ are the decision variables. We call λ, ν , and Z the *inner-dual variables*. By weak duality of convex SDPs, every feasible point of (12) upper-bounds the SDP value in (11), and therefore

$$\mathcal{R}(M) \leq \overline{\mathcal{R}}(M).$$

Consequently, to prove a convergence guarantee it suffices to exhibit a single feasible point of (12): its objective value νR^2 is a valid upper bound on the worst-case performance. Strong duality is never needed for this direction.

Finally, we record the scale invariance of the setup [11, §3.5]: it suffices to work with $L = 1$ and $R = 1$, since for any other $L > 0$ and $R > 0$ the stepsizes h and α are unchanged and the worst-case value scales as $L^2 R^2$.

6 The analytic certificate

We now state the analytic solution: the stepsizes of the method together with inner-dual variables that, as Section 7 proves, form a feasible point of (12). How these expressions were originally found is described in Remark 1; the verification below is entirely independent of that discovery process. The expressions are stated in terms of Ω_N and $\{\rho_n\}_{n=0}^{N+1}$, which satisfy Lemma 1.

For the stepsizes,

$$h_{i,j} = \delta_{i,j+1} + \Omega_N \left(\frac{1 + \rho_{N-i+1}^2}{2\rho_{N-i+1}} - \frac{1 + \rho_{N-i}^2}{2\rho_{N-i}} \right) \left(\frac{1 + \rho_{j+1}^2}{2\rho_{j+1}} - \frac{1 + \rho_j^2}{2\rho_j} \right), \quad 0 \leq j \leq i-1, \quad (13)$$

or equivalently

$$\alpha_{i,j} = \Omega_N \left((\rho_j - \rho_{j+1}) - \left(\frac{1 + \rho_{j+1}^2}{\rho_{j+1}} - \frac{1 + \rho_j^2}{\rho_j} \right) \frac{\rho_{i+1}^2}{1 - \rho_{i+1}^2} \right), \quad 0 \leq j \leq i-1. \quad (14)$$

For inner-dual variables of (12) for the FSFOM M given by (14), $\nu = 1/\Omega_N^2$ and

$$\lambda = \begin{cases} \lambda_{i,i+1} = \frac{1}{\Omega_N} \frac{1 - \rho_{i+1}^2}{\rho_{i+1}}, & 0 \leq i \leq N-1, \\ \lambda_{\star,i} = \frac{2}{\Omega_N} (\rho_i - \rho_{i+1}), & 0 \leq i \leq N, \\ \lambda_{N,i} = \frac{1}{\Omega_N} \left(\frac{1 + \rho_{i+1}^2}{\rho_{i+1}} - \frac{1 + \rho_i^2}{\rho_i} \right), & 0 \leq i \leq N-1, \\ \lambda_{N,\star} = \frac{2}{\Omega_N}, \\ \lambda_{i,j} = 0, & \text{otherwise,} \end{cases} \quad (15)$$

and

$$Z = \nu B_{0,\star} - C_{N,\star} + \sum_{i,j \in I_N^* : i \neq j} \lambda_{i,j} \left[A_{i,j}(\alpha) + \frac{1}{2L} C_{i,j} \right]. \quad (16)$$

7 Feasibility of the certificate

In this section, we prove the convergence statement of Theorem 1. By the PEP machinery developed in the previous section, it suffices to prove the solution provided by (13)–(16) is feasible for the dual inner problem (12). By the scaling argument stated at the end of Section 5, we assume $L = R = 1$.

Introduction of auxiliary variables. We define a few parameters that will be useful for simplifying (13)–(16):

$$\begin{aligned}
a_i &\triangleq \Omega_N \frac{1 - \rho_i^2}{2\rho_i}, \quad 0 \leq i \leq N, \\
b_i &\triangleq \Omega_N (\rho_i - \rho_{i+1}), \quad 0 \leq i \leq N, \\
c_i &\triangleq \Omega_N \left(\frac{1 + \rho_{i+1}^2}{2\rho_{i+1}} - \frac{1 + \rho_i^2}{2\rho_i} \right), \quad 0 \leq i \leq N-1, \\
d &\triangleq \Omega_N, \\
r_i &\triangleq \frac{2\rho_i^2}{1 - \rho_i^2}, \quad 1 \leq i \leq N+1.
\end{aligned} \tag{17}$$

Using the terms in (17), the stepsize (13) can be simplified as

$$h_{i,j} = \delta_{i,j+1} + \frac{1}{\Omega_N} c_j c_{N-i}, \quad 0 \leq j < i \leq N.$$

We construct a matrix λ from the entries of $\{\lambda_{i,j}\}_{i,j \in I_N^*: i \neq j}$ such that the rows and columns are associated with the indices I_N^* with the diagonal entries being zero. Denote by $\tilde{\lambda}$ the normalized matrix of dual multipliers λ , which is given by the following, where the first row denotes $\lambda_{\star,j}$, and the last row denotes $\lambda_{N,j}$:

$$\lambda = 2\nu \times \begin{pmatrix} 0 & b_0 & b_1 & \dots & b_{N-1} & b_N \\ & 0 & a_1 & & & \\ & & 0 & a_2 & & \\ & & & \ddots & \ddots & \\ & & & & 0 & a_N \\ d & c_0 & c_1 & \dots & c_{N-1} & 0 \end{pmatrix} = 2\nu \times \tilde{\lambda}. \tag{18}$$

Lemma 4. *The following equality holds for auxiliary variables (note that $a_0 = 0$ since $\rho_0 = 1$):*

$$\begin{aligned}
a_{i+1} &= b_i c_i + \frac{1}{2} b_i^2, \quad 0 \leq i \leq N-1, \\
c_{N-i} &= \Omega_N (r_i - r_{i+1}), \quad 1 \leq i \leq N.
\end{aligned} \tag{19}$$

Moreover, the balance identities

$$\begin{aligned}
a_i + b_i + c_i &= a_{i+1}, \quad 0 \leq i \leq N-1, \\
d + \sum_{i=0}^{N-1} c_i &= a_N + b_N, \\
\sum_{i=0}^N b_i &= d
\end{aligned} \tag{20}$$

hold, together with

$$a_i r_i = \Omega_N \rho_i, \quad 1 \leq i \leq N. \tag{21}$$

In particular, $a_N r_N = 1 = b_N$.

Proof. We prove the first line of (19) by showing its equivalence to the recurrence relation (3):

$$\begin{aligned} a_{i+1} &= b_i c_i + \frac{1}{2} b_i^2 \\ \Leftrightarrow \frac{1 - \rho_{i+1}^2}{2\rho_{i+1}} &= \Omega_N(\rho_i - \rho_{i+1}) \left(\frac{1}{2\rho_{i+1}} - \frac{1}{2\rho_i} \right) \\ \Leftrightarrow \rho_i(1 - \rho_{i+1}^2) &= \Omega_N(\rho_i - \rho_{i+1})^2. \end{aligned}$$

Using the relation $\rho_{N+1-i} = T(\rho_i) = (1 - \rho_i)/(1 + \rho_i)$, for any $1 \leq i \leq N + 1$, we have

$$\frac{1 + \rho_{N-i+1}^2}{2\rho_{N-i+1}} = \frac{1 + T(\rho_i)^2}{2T(\rho_i)} = \frac{1 + \rho_i^2}{1 - \rho_i^2} = r_i + 1.$$

Hence, for any $1 \leq i \leq N$, we have

$$c_{N-i} = \Omega_N \left(\frac{1 + \rho_{N-i+1}^2}{2\rho_{N-i+1}} - \frac{1 + \rho_{N-i}^2}{2\rho_{N-i}} \right) = \Omega_N(r_i - r_{i+1}),$$

which proves the second line of (19).

For the balance identities (20), telescoping gives

$$\sum_{i=0}^N b_i = \Omega_N(\rho_0 - \rho_{N+1}) = \Omega_N = d,$$

and for $0 \leq i \leq N - 1$, using $a_0 = 0$,

$$a_i + b_i + c_i = \Omega_N \left(\frac{1 - \rho_i^2}{2\rho_i} + \rho_i - \rho_{i+1} + \frac{1 + \rho_{i+1}^2}{2\rho_{i+1}} - \frac{1 + \rho_i^2}{2\rho_i} \right) = \Omega_N \frac{1 - \rho_{i+1}^2}{2\rho_{i+1}} = a_{i+1}.$$

Combining the two,

$$d + \sum_{i=0}^{N-1} c_i = \sum_{i=0}^{N-1} (b_i + c_i) + b_N = \sum_{i=0}^{N-1} (a_{i+1} - a_i) + b_N = a_N + b_N.$$

Finally, (21) follows directly from the definitions (17):

$$a_i r_i = \Omega_N \frac{1 - \rho_i^2}{2\rho_i} \cdot \frac{2\rho_i^2}{1 - \rho_i^2} = \Omega_N \rho_i,$$

and at $i = N$ we get $a_N r_N = \Omega_N \rho_N = 1 = b_N$ from $\rho_N = 1/\Omega_N$. □

Equivalence of the two stepsize forms. For $0 \leq j \leq i \leq N$, define the extended stepsizes $\bar{\alpha}_{i,j}$ as $\bar{\alpha}_{i,j} \triangleq \alpha_{i,j} = \sum_{k=j+1}^i h_{k,j}$ for $j < i$ and $\bar{\alpha}_{i,i} \triangleq 1$ for $0 \leq i \leq N$.

With this convention,

$$h_{i,j} = \bar{\alpha}_{i,j} - \bar{\alpha}_{i-1,j} + \delta_{i,j+1}, \quad 0 \leq j < i \leq N.$$

The following lemma proves that (13) leads to (14).

Lemma 5. For $0 \leq j \leq i \leq N$, $(i, j) \neq (N, N)$,

$$\bar{\alpha}_{i,j} = b_j - c_j r_{i+1}. \quad (22)$$

Moreover,

$$\bar{\alpha}_{N,j} = b_j, \quad 0 \leq j \leq N.$$

Proof. Recalling the definition (13) of $h_{i,j}$, and using the second line of (19), we get for $0 \leq j < i \leq N$,

$$\bar{\alpha}_{i,j} = \sum_{k=j+1}^i \left(\delta_{k,j+1} + \frac{1}{\Omega_N} c_j c_{N-k} \right) = 1 + c_j \sum_{k=j+1}^i (r_k - r_{k+1}) = 1 + c_j (r_{j+1} - r_{i+1}). \quad (23)$$

For $i = j < N$, $\bar{\alpha}_{i,i} = 1 = 1 + c_i (r_{i+1} - r_{i+1})$, so (23) also holds for this case.

Since $r_{N+1} = 0$, for $i = N$, $0 \leq j \leq N-1$, we have

$$\begin{aligned} \bar{\alpha}_{N,j} &= 1 + c_j r_{j+1} = 1 + \Omega_N \left(\frac{1 + \rho_{j+1}^2}{2\rho_{j+1}} - \frac{1 + \rho_j^2}{2\rho_j} \right) \frac{2\rho_{j+1}^2}{1 - \rho_{j+1}^2} \\ &= 1 + \Omega_N \left(\rho_j - \rho_{j+1} - \frac{1}{\Omega_N} \right) = b_j, \end{aligned}$$

where the middle step is justified by

$$\begin{aligned} &\left(\rho_{j+1} + \frac{1}{\rho_{j+1}} - \rho_j - \frac{1}{\rho_j} \right) \rho_{j+1}^2 = (1 - \rho_{j+1}^2) \left(\rho_j - \rho_{j+1} - \frac{1}{\Omega_N} \right) \\ \Leftrightarrow &\rho_{j+1} - \frac{\rho_{j+1}^2}{\rho_j} = \rho_j - \rho_{j+1} - \frac{1}{\Omega_N} + \frac{\rho_{j+1}^2}{\Omega_N} \\ \Leftrightarrow &\rho_{j+1}(\rho_j - \rho_{j+1}) = \rho_j \left(\rho_j - \rho_{j+1} - \frac{1}{\Omega_N} + \frac{\rho_{j+1}^2}{\Omega_N} \right) \\ \Leftrightarrow &\Omega_N (\rho_j - \rho_{j+1})^2 = \rho_j (1 - \rho_{j+1}^2), \end{aligned}$$

which is the original recurrence relation (3).

For the case $i = j = N$, using $b_N = \Omega_N(1/\Omega_N - 0) = 1$, $\bar{\alpha}_{N,N} = b_N$ also holds.

Substituting into (23) gives, for $0 \leq j \leq i \leq N$, $(i, j) \neq (N, N)$,

$$\bar{\alpha}_{i,j} = 1 + c_j (r_{j+1} - r_{i+1}) = \bar{\alpha}_{N,j} - c_j r_{i+1} = b_j - c_j r_{i+1}.$$

□

If we consider $0 \leq j < i \leq N$, we recover (14):

$$\alpha_{i,j} = \bar{\alpha}_{i,j} = b_j - c_j r_{i+1} = \Omega_N \left(\left(\rho_j - \rho_{j+1} \right) - \left(\frac{1 + \rho_{j+1}^2}{\rho_{j+1}} - \frac{1 + \rho_j^2}{\rho_j} \right) \frac{\rho_{i+1}^2}{1 - \rho_{i+1}^2} \right).$$

Feasibility of the certificate. For ease of calculation, we define $\tilde{\lambda} \triangleq \lambda/(2\nu)$ and $\tilde{Z} \triangleq Z/\nu$. We prove the feasibility by checking the constraints of (12). The second constraint holds by the definition of Z (16), so we check the other three constraints.

1. Nonnegativity of the dual multipliers ν and λ .

The multiplier $\nu = 1/\Omega_N^2 > 0$ is strictly positive. For λ note that since $1 = \rho_0 > \rho_1 > \dots > \rho_N > \rho_{N+1} = 0$, we have $a_i > 0$ for $1 \leq i \leq N$, $b_i > 0$ for $0 \leq i \leq N$, and $d = \Omega_N > 0$. Moreover, the map $x \mapsto (1 + x^2)/(2x)$ is strictly decreasing on $(0, 1)$, so $c_i > 0$ for $0 \leq i \leq N - 1$. Thus λ written as in (18) is nonnegative.

2. The linear constraint $\sum_{i,j \in I_N^*: i \neq j} \lambda_{i,j} (\mathbf{f}_j - \mathbf{f}_i) = \mathbf{0}$.

Substituting the dual multipliers (18) and $\mathbf{f}_\star = \mathbf{0}$, and dividing by $2\nu > 0$, the linear constraint becomes

$$\sum_{i=1}^N a_i (\mathbf{f}_i - \mathbf{f}_{i-1}) + \sum_{j=0}^N b_j \mathbf{f}_j + \sum_{j=0}^{N-1} c_j (\mathbf{f}_j - \mathbf{f}_N) - d \mathbf{f}_N = \mathbf{0}.$$

Since the selectors $\mathbf{f}_0, \dots, \mathbf{f}_N$ are linearly independent, this holds if and only if the coefficient of each $\mathbf{f}_k$ vanishes. By the balance identities (20) of Lemma 4, the coefficient of $\mathbf{f}_0$ is $-a_1 + b_0 + c_0 = 0$ (the first identity at $i = 0$, with $a_0 = 0$); the coefficient of $\mathbf{f}_k$ for $1 \leq k \leq N - 1$ is $a_k - a_{k+1} + b_k + c_k = 0$ (the same identity at $i = k$); and the coefficient of $\mathbf{f}_N$ is $a_N + b_N - \sum_{j=0}^{N-1} c_j - d = 0$ (the second identity).

3. Positive semidefiniteness of Z .

Lemma 6. *The dual slack matrix Z defined in (16) can be factored as*

$$Z = \nu \times \begin{bmatrix} 1 \\ -b_0 \\ \vdots \\ -b_N \end{bmatrix} \begin{bmatrix} 1 \\ -b_0 \\ \vdots \\ -b_N \end{bmatrix}^\top = \nu \times \tilde{Z}.$$

Since $\nu > 0$, it follows that $Z \succeq 0$.

Proof. Using the definitions (10), the normalized dual slack matrix $\tilde{Z}$ can be written as

$$\begin{aligned} \tilde{Z} &= B_{0,\star} - \frac{1}{\nu} C_{N,\star} + 2 \sum_{i,j \in I_N^*: i \neq j} \frac{\lambda_{i,j}}{2\nu} \left[A_{i,j}(\alpha) + \frac{1}{2} C_{i,j} \right] \\ &= ((\mathbf{x}_0 - \mathbf{x}_\star) \odot (\mathbf{x}_0 - \mathbf{x}_\star)) - \frac{1}{\nu} ((\mathbf{g}_N - \mathbf{g}_\star) \odot (\mathbf{g}_N - \mathbf{g}_\star)) \\ &\quad + 2 \sum_{i,j \in I_N^*: i \neq j} \tilde{\lambda}_{i,j} \left[(\mathbf{g}_j \odot (\mathbf{x}_i - \mathbf{x}_j)) + \frac{1}{2} ((\mathbf{g}_i - \mathbf{g}_j) \odot (\mathbf{g}_i - \mathbf{g}_j)) \right]. \end{aligned}$$

Putting the value of (18) and $\nu = 1/\Omega_N^2$ on the right hand side, we get the following expression:

$$\begin{aligned} \tilde{Z} &= -\Omega_N^2 ((\mathbf{g}_N - \mathbf{g}_\star) \odot (\mathbf{g}_N - \mathbf{g}_\star)) + ((\mathbf{x}_0 - \mathbf{x}_\star) \odot (\mathbf{x}_0 - \mathbf{x}_\star)) \\ &\quad + 2 \sum_{i=1}^N a_i \left((\mathbf{g}_i \odot (\mathbf{x}_{i-1} - \mathbf{x}_i)) + \frac{1}{2} ((\mathbf{g}_i - \mathbf{g}_{i-1}) \odot (\mathbf{g}_i - \mathbf{g}_{i-1})) \right) \\ &\quad + 2 \sum_{i=0}^N b_i \left(-(\mathbf{g}_i \odot \mathbf{x}_i) + \frac{1}{2} (\mathbf{g}_i \odot \mathbf{g}_i) \right) \end{aligned}$$

$$\begin{aligned}
& + 2 \sum_{i=0}^{N-1} c_i \left((\mathbf{g}_i \odot (\mathbf{x}_N - \mathbf{x}_i)) + \frac{1}{2} ((\mathbf{g}_i - \mathbf{g}_N) \odot (\mathbf{g}_i - \mathbf{g}_N)) \right) \\
& + d(\mathbf{g}_N \odot \mathbf{g}_N) \\
& = (\mathbf{x}_0 \odot \mathbf{x}_0) - 2 \sum_{i=0}^N b_i (\mathbf{g}_i \odot \mathbf{x}_0) + S,
\end{aligned}$$

where we have used $\mathbf{x}_\star = 0$ and $\mathbf{g}_\star = 0$ and have defined a matrix

$$\begin{aligned}
S & \triangleq -\Omega_N^2 (\mathbf{g}_N \odot \mathbf{g}_N) + 2 \sum_{i=1}^N a_i \left((\mathbf{g}_i \odot (\mathbf{x}_{i-1} - \mathbf{x}_i)) + \frac{1}{2} ((\mathbf{g}_i - \mathbf{g}_{i-1}) \odot (\mathbf{g}_i - \mathbf{g}_{i-1})) \right) \\
& + 2 \sum_{i=0}^N b_i \left(-(\mathbf{g}_i \odot (\mathbf{x}_i - \mathbf{x}_0)) + \frac{1}{2} (\mathbf{g}_i \odot \mathbf{g}_i) \right) \\
& + 2 \sum_{i=0}^{N-1} c_i \left((\mathbf{g}_i \odot (\mathbf{x}_N - \mathbf{x}_i)) + \frac{1}{2} ((\mathbf{g}_i - \mathbf{g}_N) \odot (\mathbf{g}_i - \mathbf{g}_N)) \right) \\
& + d(\mathbf{g}_N \odot \mathbf{g}_N)
\end{aligned}$$

for ease of simplification. Since

$$\begin{aligned}
\begin{bmatrix} 1 \\ -b_0 \\ \vdots \\ -b_N \end{bmatrix} \begin{bmatrix} 1 \\ -b_0 \\ \vdots \\ -b_N \end{bmatrix}^\top & = (\mathbf{x}_0 - \sum_{i=0}^N b_i \mathbf{g}_i) \odot (\mathbf{x}_0 - \sum_{i=0}^N b_i \mathbf{g}_i) \\
& = (\mathbf{x}_0 \odot \mathbf{x}_0) - 2 \sum_{i=0}^N b_i (\mathbf{g}_i \odot \mathbf{x}_0) + \sum_{i=0}^N \sum_{j=0}^N b_i b_j (\mathbf{g}_i \odot \mathbf{g}_j),
\end{aligned}$$

it remains to prove $S = \sum_{i=0}^N \sum_{j=0}^N b_i b_j (\mathbf{g}_i \odot \mathbf{g}_j)$.

Recall that expressing the differences using $\bar{\alpha}_{i,j}$, we have for any $0 \leq i \leq N$,

$$\begin{aligned}
\mathbf{x}_0 - \mathbf{x}_i & = \sum_{j=1}^i \sum_{k=0}^{j-1} h_{j,k} \mathbf{g}_k = \sum_{k=0}^{i-1} \bar{\alpha}_{i,k} \mathbf{g}_k, \\
\mathbf{x}_i - \mathbf{x}_N & = (\mathbf{x}_0 - \mathbf{x}_N) - (\mathbf{x}_0 - \mathbf{x}_i) = \sum_{k=0}^{N-1} \bar{\alpha}_{N,k} \mathbf{g}_k - \sum_{k=0}^{i-1} \bar{\alpha}_{i,k} \mathbf{g}_k.
\end{aligned}$$

Then, the remaining part of the slack matrix is calculated as

$$\begin{aligned}
S & = -\Omega_N^2 (\mathbf{g}_N \odot \mathbf{g}_N) + \sum_{i=1}^N a_i ((\mathbf{g}_i - \mathbf{g}_{i-1}) \odot (\mathbf{g}_i - \mathbf{g}_{i-1})) + 2 \sum_{i=1}^N \sum_{j=0}^{i-1} a_i h_{i,j} (\mathbf{g}_i \odot \mathbf{g}_j) \\
& + \sum_{i=0}^N b_i (\mathbf{g}_i \odot \mathbf{g}_i) + 2 \sum_{i=0}^N \sum_{j=0}^{i-1} b_i \bar{\alpha}_{i,j} (\mathbf{g}_i \odot \mathbf{g}_j) \\
& + \sum_{i=0}^{N-1} c_i ((\mathbf{g}_i - \mathbf{g}_N) \odot (\mathbf{g}_i - \mathbf{g}_N)) - 2 \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} c_i \bar{\alpha}_{N,j} (\mathbf{g}_i \odot \mathbf{g}_j)
\end{aligned}$$

$$+ 2 \sum_{i=0}^{N-1} \sum_{j=0}^{i-1} c_i \bar{\alpha}_{i,j} (\mathbf{g}_i \odot \mathbf{g}_j) + d (\mathbf{g}_N \odot \mathbf{g}_N).$$

We prove $S = \sum_{i=0}^N \sum_{j=0}^N b_i b_j (\mathbf{g}_i \odot \mathbf{g}_j)$ by equating the matrix entries explicitly; since $\mathbf{g}_i = e_{i+2}$, it suffices to prove that the $(i+2, j+2)$ -th entry of S is equal to $b_i b_j$ for $0 \leq i, j \leq N$.

Off-diagonal entries.

Since S is a symmetric matrix, it suffices to calculate only the entries $S_{i+2, j+2}$ where $i \geq j$.

For $0 \leq j < i < N$,

$$\begin{aligned} S_{i+2, j+2} &= -a_i \delta_{i, j+1} + a_i h_{i, j} + b_i \bar{\alpha}_{i, j} - c_i \bar{\alpha}_{N, j} - c_j \bar{\alpha}_{N, i} + c_i \bar{\alpha}_{i, j} \\ &= -a_i \delta_{i, j+1} + a_i (\bar{\alpha}_{i, j} - \bar{\alpha}_{i-1, j} + \delta_{i, j+1}) + b_i \bar{\alpha}_{i, j} - c_i b_j - c_j b_i + c_i \bar{\alpha}_{i, j} \\ &= (a_i + b_i + c_i) \bar{\alpha}_{i, j} - a_i \bar{\alpha}_{i-1, j} - c_i b_j - c_j b_i \\ &= a_{i+1} \bar{\alpha}_{i, j} - a_i \bar{\alpha}_{i-1, j} - c_i b_j - c_j b_i \\ &= a_{i+1} (b_j - c_j r_{i+1}) - a_i (b_j - c_j r_i) - c_i b_j - c_j b_i \\ &= (a_{i+1} - a_i) b_j + c_j (a_i r_i - a_{i+1} r_{i+1}) - c_i b_j - c_j b_i \\ &= (b_i + c_i) b_j + c_j b_i - c_i b_j - c_j b_i \\ &= b_i b_j, \end{aligned}$$

where we used Lemma 5 in the fifth equality, the balance identities (20) in the fourth and seventh, and, also in the seventh, $a_i r_i - a_{i+1} r_{i+1} = \Omega_N (\rho_i - \rho_{i+1}) = b_i$, which follows from (21).

For $i = N$, $0 \leq j \leq N-1$, we have

$$\begin{aligned} S_{N+2, j+2} &= -a_N \delta_{N, j+1} + a_N h_{N, j} + b_N \bar{\alpha}_{N, j} - c_j \\ &= a_N (h_{N, j} - \delta_{N, j+1}) + b_N b_j - c_j \\ &= a_N \left(\frac{1}{\Omega_N} c_j c_0 \right) + b_N b_j - c_j \\ &= a_N (c_j r_N) + b_N b_j - c_j \\ &= c_j b_N + b_N b_j - c_j \\ &= b_N b_j, \end{aligned}$$

where the fourth equality uses $c_0 = \Omega_N r_N$ (the second line of (19) at $i = N$, with $r_{N+1} = 0$), the fifth equality uses $a_N r_N = 1 = b_N$ from (21), and the last uses $b_N = 1$.

Diagonal entries.

For $0 \leq i \leq N-1$,

$$\begin{aligned} S_{i+2, i+2} &= a_i + a_{i+1} + b_i + c_i - 2c_i \bar{\alpha}_{N, i} \\ &= 2a_{i+1} - 2c_i b_i = b_i^2, \end{aligned}$$

where the last equation comes from the first line of (19).

For $i = N$, we have

$$\begin{aligned} S_{N+2, N+2} &= -\Omega_N^2 + a_N + b_N + \sum_{i=0}^{N-1} c_i + d \\ &= -\Omega_N^2 + a_N + b_N + (a_N + b_N) = -\Omega_N^2 + 2 + 2a_N = 1, \end{aligned}$$

where we have used the balance identities (20), $\rho_N = 1/\Omega_N$, and

$$a_N = \Omega_N \frac{1 - \rho_N^2}{2\rho_N} = \frac{\Omega_N^2 - 1}{2}.$$

Since $b_N = 1$, we have $S_{N+2, N+2} = 1 = b_N^2$. □

Therefore, (13)–(16) is feasible for the problem (12).

In the normalized setting $L = R = 1$, the certificate above is feasible for (12) with objective value $\nu R^2 = 1/\Omega_N^2$, so weak duality gives $\mathcal{R}(M) \leq \overline{\mathcal{R}}(M) \leq 1/\Omega_N^2$ for the lemniscate acceleration M . Undoing the normalization by the scaling argument stated at the end of Section 5 and choosing $R = \|x_0 - x_\star\|$, we conclude that $\|\nabla f(x_N)\|^2 \leq L^2 \|x_0 - x_\star\|^2 / \Omega_N^2$ for every L -smooth convex f with a minimizer $x_\star$, which is the first inequality of Theorem 1; the second inequality is the bound $\Omega_N > (N+1)^2/\varpi^2$ of Corollary 1. It only remains to show that the FSFOM defined by the stepsize matrix (13) is equivalent to (LemniAcc), which is proved in the following Corollary 2.

8 Equivalence of the method to (LemniAcc)

We state and prove the equivalence of three representations of a fixed-step first-order method: the h -matrix form, the memory-efficient form, and the momentum form.

Lemma 7 (Equivalence of representations). *Let $v, w \in \mathbb{R}^N$ be vectors with $v_k \neq 0$. Let M be a fixed step first-order method with stepsize matrix $h_{i,j} = \delta_{i,j+1} + v_i w_j$ for $0 \leq j < i \leq N$, defined by*

$$x_{k+1} = x_k - \sum_{j=0}^k h_{k+1,j} \frac{1}{L} \nabla f(x_j), \quad 0 \leq k \leq N-1.$$

Then, M has a memory-efficient representation

$$\begin{aligned} z_{k+1} &= z_k - w_k \frac{1}{L} \nabla f(x_k), \quad 0 \leq k \leq N-1, \\ x_{k+1} &= x_k^+ + v_{k+1} z_{k+1}, \end{aligned}$$

where $x_k^+ = x_k - (1/L) \nabla f(x_k)$ and $z_0 = 0$. Also, M has a momentum form

$$x_{k+1} = \begin{cases} x_0^+ + \eta_1 (x_0^+ - x_0), & k = 0, \\ x_k^+ + \zeta_{k+1} (x_k^+ - x_{k-1}^+) + \eta_{k+1} (x_k^+ - x_k), & 1 \leq k \leq N-1, \end{cases}$$

with coefficients $\eta_1 = v_1 w_0$ and, for $1 \leq k \leq N-1$, $\zeta_{k+1} = v_{k+1}/v_k$ and $\eta_{k+1} = v_{k+1}(w_k - 1/v_k)$. In particular, the memory-efficient representation and the momentum form are equivalent to each other: started from the same x_0 (with $z_0 = 0$), they generate the same iterates $x_0, \dots, x_N$.

Proof. Denote $g_k \triangleq (1/L) \nabla f(x_k)$, so that $x_k^+ = x_k - g_k$.

Memory-efficient form. Set $z_{k+1} \triangleq -\sum_{j=0}^k w_j g_j$, which starts at $z_0 = 0$ (empty sum) and satisfies $z_{k+1} = z_k - w_k g_k$. Since $\delta_{k+1,j+1} = \delta_{k,j}$, splitting the update of M along $h_{k+1,j} = \delta_{k,j} + v_{k+1} w_j$ gives

$$x_{k+1} = x_k - \sum_{j=0}^k (\delta_{k,j} + v_{k+1} w_j) g_j = x_k^+ + v_{k+1} z_{k+1},$$

which is the memory-efficient recursion.

Momentum form. Fix $1 \leq k \leq N-1$. The memory recursion at the previous step reads $x_k = x_{k-1}^+ + v_k z_k$, so $z_k = (x_k - x_{k-1}^+)/v_k$ (using $v_k \neq 0$). Substituting this and $g_k = x_k - x_k^+$ into $z_{k+1} = z_k - w_k g_k$ and then into $x_{k+1} = x_k^+ + v_{k+1} z_{k+1}$ gives

$$x_{k+1} = x_k^+ + v_{k+1} \left(\frac{x_k - x_{k-1}^+}{v_k} - w_k (x_k - x_k^+) \right);$$

regrouping via $x_k - x_{k-1}^+ = (x_k^+ - x_{k-1}^+) + (x_k - x_k^+)$ and collecting coefficients, a routine calculation, yields the momentum form with $\zeta_{k+1} = v_{k+1}/v_k$ and $\eta_{k+1} = v_{k+1}(w_k - 1/v_k)$. The first step is separate: $z_1 = -w_0 g_0$ gives $x_1 = x_0^+ - v_1 w_0 g_0 = x_0^+ + \eta_1 (x_0^+ - x_0)$ with $\eta_1 = v_1 w_0$. $\square$

We now show the equivalence of recursion (LemniAcc) and its momentum form (2) with the h -matrix representation (13) as an instance of Lemma 7.

Corollary 2. *The method (LemniAcc) and its momentum form (2) are equivalent to the h -matrix representation (13).*

Proof. Write $\phi_n \triangleq (1 + \rho_n^2)/(2\rho_n)$. Direct substitution of the symmetry $\rho_{N+1-k} = (1 - \rho_k)/(1 + \rho_k)$ noted in Section 3 gives $\phi_{N+1-k} = (1 + \rho_k^2)/(1 - \rho_k^2)$ for $1 \leq k \leq N+1$, so the x -update coefficient of (LemniAcc) satisfies

$$\frac{1 + \rho_{k+1}^2}{1 - \rho_{k+1}^2} - \frac{1 + \rho_{k+2}^2}{1 - \rho_{k+2}^2} = \phi_{N-k} - \phi_{N-k-1}.$$

Hence (LemniAcc) is the memory-efficient representation of Lemma 7 with $v_i = \phi_{N-i+1} - \phi_{N-i}$ and $w_j = \Omega_N (\phi_{j+1} - \phi_j)$, where ϕ_n is strictly monotone in n so that $v_i \neq 0$, and $h_{i,j} = \delta_{i,j+1} + v_i w_j$ is exactly (13); the claim then follows from Lemma 7. We note that the rank-one factor of (13) determines (v, w) only up to the rescaling $(v, w) \mapsto (cv, w/c)$ with $c \neq 0$; the symmetric choice $v_i = \sqrt{\Omega_N} (\phi_{N-i+1} - \phi_{N-i})$, $w_j = \sqrt{\Omega_N} (\phi_{j+1} - \phi_j)$ satisfies $w_{N-i} = v_i$ and thereby exhibits a time-reversal symmetry of (LemniAcc). $\square$

This completes the proof of Theorem 1.

9 How the certificate was found

Remark 1. The certificate verified in this document was not constructed by hand. Following the branch-and-bound performance estimation programming (BnB-PEP) methodology [2], the stepsizes are treated as decision variables alongside the inner-dual variables: substituting the dual (12) for the inner maximization turns the design of an optimal method into a single minimization problem, jointly over α and (λ, ν, Z) , and replacing the semidefinite constraint $Z \succeq 0$ with the Cholesky factorization $Z = VV^\top$, where V is lower triangular with nonnegative diagonals [4, Corollary 7.2.9], makes the joint problem a nonconvex quadratically constrained quadratic program that spatial branch-and-bound solvers can address. Solving this problem numerically for small N and fitting symbolic expressions to the resulting solutions, as described in the companion paper [1], produced the analytic formulas of Section 6, including the recurrence of Lemma 1. The verification given in Sections 7 and 8 is entirely classical and independent of this discovery process.

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